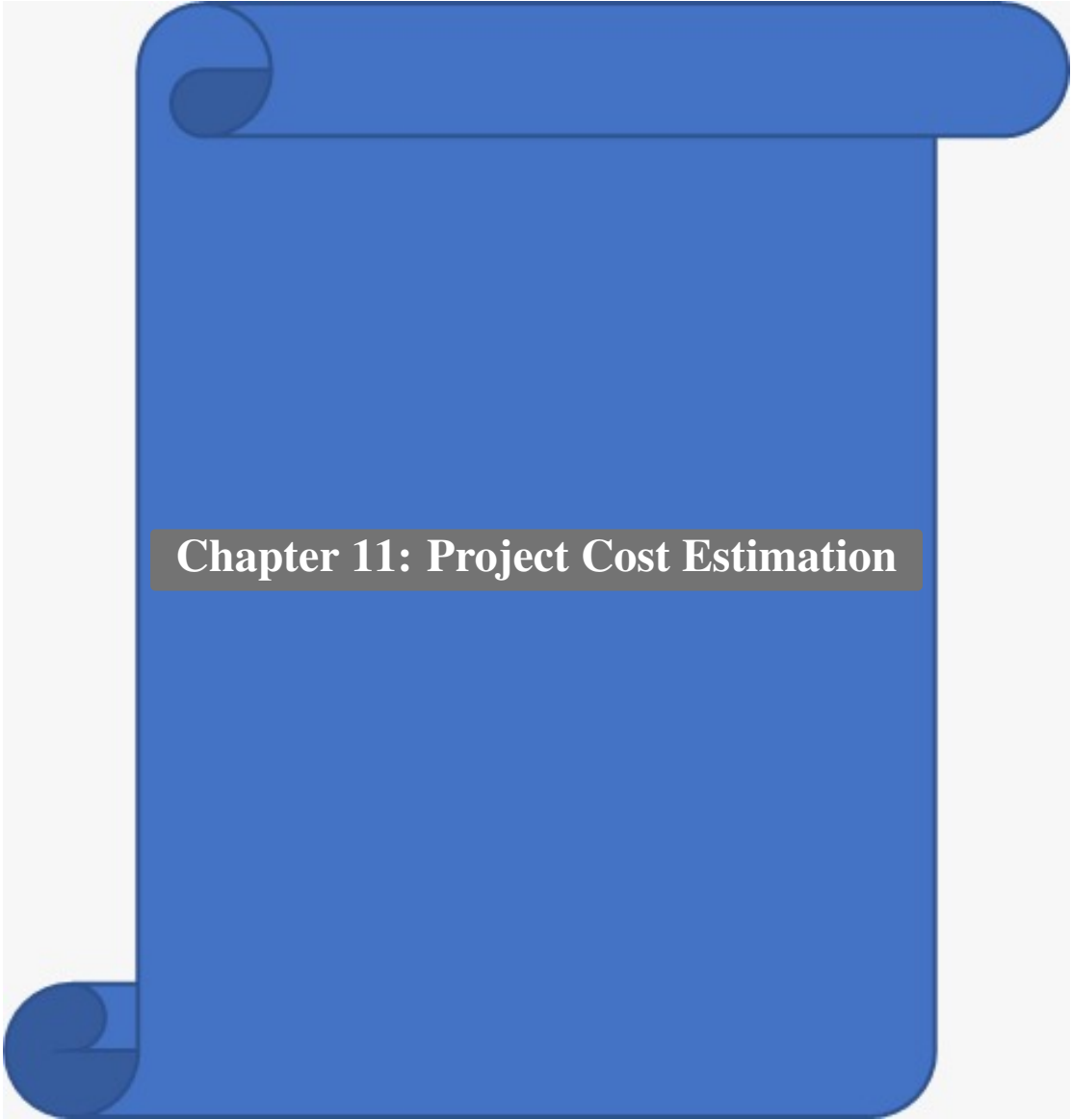




Chapter 11: Project Cost Estimation

11.1 COCOMO Model Estimation

The Constructive Cost Model (COCOMO) is used to estimate the effort, development time, and team size required for the AgriTrace project. The Basic COCOMO model uses the following formulas:

- Effort (E) = $a \times (KLOC)^b$ person-months
- Development Time (D) = $c \times (E)^d$ months
- People Required (P) = E / D

For a semi-detached software project:

- $a = 3.0, b = 1.12, c = 2.5, d = 0.35$

Estimation: The AgriTrace codebase comprises approximately 8.5 KLOC (thousand lines of code), including TypeScript source files, configuration files, and Firestore security rules.

Table 11.1: COCOMO Effort Estimation

Parameter	Formula	Value
Lines of Code (KLOC)	Measured	8.5
Effort (E)	$3.0 \times (8.5)^{1.12}$	33.2 person-months
Development Time (D)	$2.5 \times (33.2)^{0.35}$	8.3 months
People Required (P)	$33.2 / 8.3$	4.0 persons
Productivity	KLOC / E	0.26 KLOC/PM

The COCOMO estimation suggests approximately 33 person-months of effort distributed across 4 team members over 8 months. Due to the use of high-level frameworks (Next.js, Firebase) and component libraries (Radix UI, Tailwind CSS), the actual development time was compressed to approximately 3 months, demonstrating the productivity gains of modern web development tools.

11.2 Itemized Cost Breakdown

The following table provides a detailed cost estimation for the AgriTrace project:

Note: The majority of the technology stack used in AgriTrace is open-source and free, significantly reducing development costs. Firebase and Vercel offer generous free tiers that are sufficient for development and small-scale production deployments. The open-source nature of the tools also eliminates vendor lock-in, allowing future migration to alternative platforms if needed.

11.3 Resource Allocation

The following table shows the resource allocation across team members:

Table 11.2: Project Cost Estimation

Item	Description	Cost (INR)
Development Costs		
Development Hardware	Laptops for 4 developers	Existing
Internet Connection	Broadband for 4 months	4,000
Software & Services (Annual)		
Firebase Spark Plan	Free tier (sufficient for development/testing)	0
Firebase Blaze Plan	Pay-as-you-go for production	2,000–5,000
Razorpay Gateway	2% per transaction (no fixed cost)	Variable
Cloudinary	Free tier (25 credits/month)	0
GitHub Repository	Free for public repositories	0
Domain Name	Custom domain (optional)	800–1,500
Vercel Hosting	Free tier for hobby projects	0
Tools & Libraries		
Next.js, React, TypeScript	Open-source (MIT License)	0
Tailwind CSS, Radix UI	Open-source (MIT License)	0
Firebase SDK	Free (Google)	0
Google Genkit AI	Free tier available	0
Lucide React Icons	Open-source (ISC License)	0
Recharts Library	Open-source (MIT License)	0
Miscellaneous		
Report Printing	5 copies, hard binding	2,500
Pen Drive	Project softcopy & PPT	300
Total Estimated Cost		9,600–12,300

Table 11.3: Resource Allocation

Team Member	Primary Role	Responsibilities
Member 1	Frontend Developer	Dashboard UIs, responsive design, Tailwind CSS styling
Member 2	Backend Developer	Firebase services, API routes, Razorpay integration
Member 3	Full-Stack Developer	Authentication, location services, notifications
Member 4	Testing & Documentation	Test cases, security rules, project report